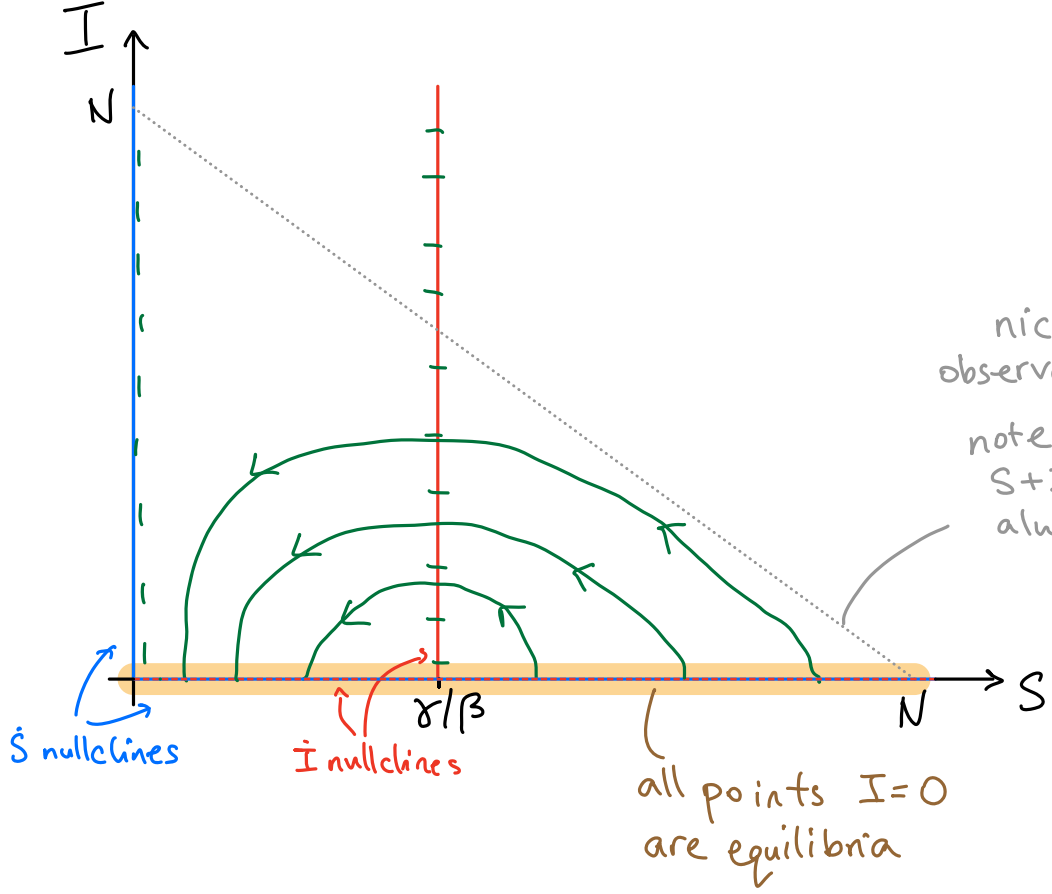




nice
observation:
note that
 $S+I \leq N$
always